Introduction to Big Data Systems: Homework 1

法兰西思科
Romero Contreras Francisco Diego Ariel

September 2025

Assumptions:

- Fixed power budget
- Total of power is proportional to the square of frequency
- Performance is proportional to frequency
- 10% of the program is sequential and 90% can be parallelism.

Solution:
We have from the assumptions that the power for the single-core case is:

$$P = c \cdot f_1 2$$

and the performance is:

$$T_1 = k \cdot f_1$$

where

- f_1 is the frequency of sequential work
- c is a constant
- k is a constant

For the N-core case, the power is:

$$P = c \cdot (0.1 \cdot f_1^2 + 0.9 \cdot N \cdot f_2^N)$$

and the performance is:

$$T_2 = k(0.1 \cdot f_1 + 0.9 \cdot f_2)$$

As the sequential process is the same as single-core, the frequency is the same for that part of the power, but the parallelism part should have another frequency.

where

- f_2 is the parallel frequency
- N the number of cores

As the Total power must be equal in both cases, we have:

$$P_{total} = c \cdot f_1^2 = c \cdot (0.1 \cdot f_1^2 + 0.9 \cdot N \cdot f_N^2)$$
$$f_1^2 = 0.1 \cdot f_1^2 + 0.9 \cdot N \cdot f_N^2$$
$$0.9 \cdot f_1^2 = 0.9 \cdot N \cdot f_N^2$$
$$f_1^2 = N \cdot f_N^2$$
$$f_1 = \sqrt{N} \cdot f_N$$
$$f_N = \frac{f_1}{\sqrt{N}}$$

Let us say that $T_1 = 1$ as an arbitrary unity, with this we calculate the speed up as the following expression:

$$SP = \frac{T_N}{T_1} = T_N$$

Later,

$$T_1 = k \cdot f_1 = 1 \Rightarrow k = \frac{1}{f_1}$$

With this we have:

$$T_N = k(0.1 \cdot f_1 + 0.9 \cdot N \cdot f_N) = \frac{1}{f_1} \cdot (0.1 \cdot f_1 + 0.9 \cdot N \cdot \frac{f_1}{\sqrt{N}})$$
$$T_N = 0.1 + 0.9 \cdot \sqrt{N}$$

So to get the speed up for dual-core, we calculate

$$T_{N=2} = 0.1 + 0.9 \cdot \sqrt{2} \approx 1.37279$$

Also, the speed up for quad-core, we calculate

$$T_{N=4} = 0.1 = 0.9 \cdot \sqrt{4} = 1.9$$